

EXPLORING THE PATHOGEN GENOMIC SURVEILLANCE SYSTEM

Outbreak-Tracing Engine — Relatedness · Clustering · Budget-Optimized Sample Prioritization

Data Structures & Algorithms – 3

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ABSTRACT

During infectious disease outbreaks, public health laboratories generate large volumes of pathogen genome sequences that must be rapidly analyzed to understand transmission dynamics and guide containment efforts. Sequencing every sample is often infeasible due to budget and turnaround-time constraints, making it critical to identify which samples carry the most epidemiological value before committing resources to full sequencing.

Relatedness Engine

- Alignment-based & alignment-free (k-mer/sketch) similarity scoring.

Transmission Network

- Graph construction, clustering, and bridge/superspreader detection.

Budget Optimization

- Greedy Set Cover for maximum-coverage sample prioritization.

Interactive Dashboard

- Relatedness matrices, cluster maps, mutation timelines, alerts.

INTRODUCTION

01

Genomic epidemiology is now central to outbreak response

Raw sequencing data alone doesn't reveal transmission structure — it needs to be turned into an actionable network.

02

Two core bottlenecks

Computational — comparing thousands of genomes efficiently; and economic — sequencing budgets are limited.

03

Existing tools solve only half the problem

Relatedness/visualization platforms (PathoSPOT, MicrobeTrace, HALviz) rarely include budget-constrained sample selection.

04

This project unifies the pipeline

Indexing → relatedness → network inference → critical-link detection → combinatorial budget optimization, in one system.

SOFTWARE REQUIREMENTS

Language

Java (core engine, graph & DSA algorithms)

Dashboard/FrontEnd

HTML,CSS,JS

Database

MongoDB

IDE

VS Code

OS

Windows

Version Control

Git & GitHub

HARDWARE REQUIREMENTS

● Processor	Intel i5 / Ryzen 5 or higher	Multi-core recommended for parallel genome comparisons
● RAM	8 GB minimum (16 GB recommended)	Larger genome datasets benefit from more headroom
● Storage	20+ GB free space	Genome sequence files + indexed k-mer data
● GPU	Optional	Not mandatory for the DSA engine; useful if scaling to ML-based classification
● Internet	Required intermittently	Pulling public datasets (NCBI, GISAID) when used

METHODOLOGY

01

Genome Indexing

k-mer based indexing for fast approximate retrieval & similarity lookup.

02

Relatedness Computation

Alignment-based scoring + alignment-free k-mer/MinHash-style comparison.

03

Mutation Detection

Diff related genome pairs; flag high-frequency mutation positions.

04

Network Construction

Build a weighted graph; apply MST to infer likely transmission paths.

05

Cluster & Bridge Detection

Partition into outbreak clusters; Tarjan's algorithm flags critical links.

06

Contact-Spread Simulation

Simulate isolating samples / breaking links; observe connectivity change.

07

Budget-Constrained Selection

Greedy Set Cover picks the minimum sample set maximizing coverage.

08

Dashboard & Alerts

Relatedness matrices, cluster maps, mutation timelines, threshold alerts.

LITERATURE SURVEY

Existing Tool	What It Does	Gap vs. This Project
PathoSPOT	Genome comparison + clonal-cluster visualization (MRSA)	No budget-optimization or sample-prioritization module
MicrobeTrace / HantaNet	Browser dashboard: networks, phylogenetic trees, maps	No coverage-optimization / Set Cover budget engine
MicroTrace (R)	SNP-based hierarchical clustering from WGS data	Clustering only — no bridge/superspreader detection
HAIViz	Interactive dashboard linking timeline, map, network	Visualization tool; no algorithmic sample prioritization
SVCollector	Greedy / ILP sample selection to maximize variant coverage	Optimizes variant discovery, not epidemiological network coverage

No surveyed tool combines relatedness scoring, bridge/superspreader detection, and Set-Cover budget optimization with a contact-spread simulator in a single pipeline — the integration is this project's key contribution.



THANK YOU

QUESTIONS · DISCUSSION · FEEDBACK

Pathogen Genomic Surveillance & Outbreak-Tracing System

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